CS598 Project 2 Report

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1 Introduction

In this report, we detail our approach to estimating the compressive strength of concrete from some measurements. We cover the shape of the data in **Exploratory Data Analysis** and each of the models we created and their performance in **Models and Performance**.

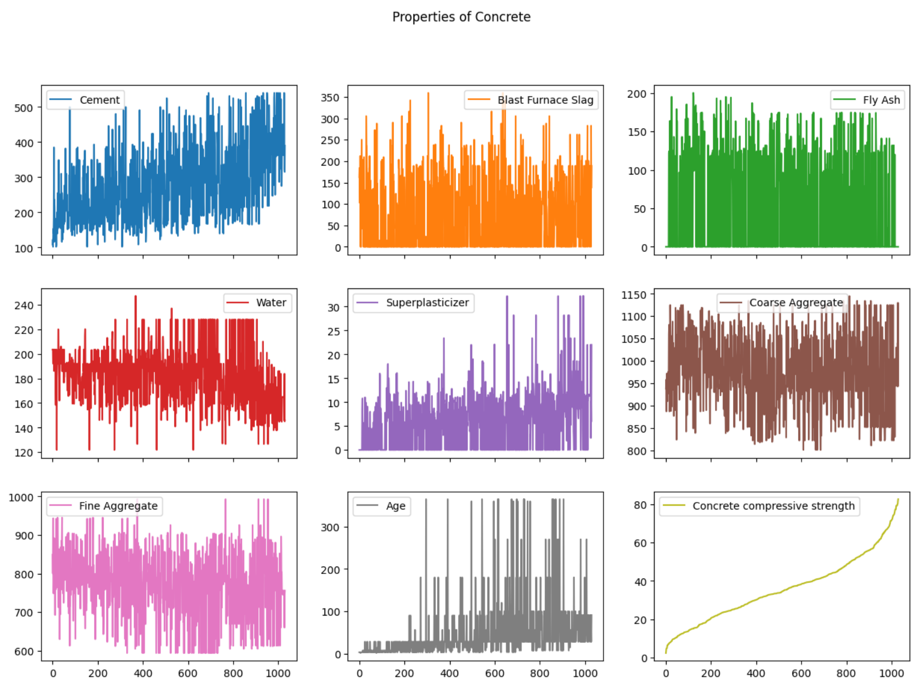


Figure 1: Graph of All Features.

2 Exploratory Data Analysis

2.1 Shape of the data

We begin by looking at a plot of each column in time in Figure 1. This plot is not meant to be looked at for a precise analysis; instead, it is used to illustrate the shape of the data.

The columns of the data have been sorted by **Concrete Compressive Strength** in ascending order. Some observations are below:

1. We note that the response variable **Concrete Compressive Strength** shows that the data contains an abundance of concrete at mid-level maturity. The sample age curve is somewhat almost linear in the middle but logarithmic/exponential at the ends. Generally, concrete integrity has a logarithmic relation to its maturity, wherein concrete cures over time to reach its mature compressive potential – meaning if the data had an even distribution of concrete at every age, the curve would look logarithmic. Since the data is graphed on a sample basis and not on a time-based axis, the shape only reflects the abundance of middle-aged concrete in the data, and that older concrete tends to mature and become stronger, but this dataset has less evidence for it.

2. We also expect splines to be especially effective with this dataset, since the response can be broken up piecewise. Because of the abundance of data in the middle age ranges and sparser data in the low and high age ranges, we can expect the regression to behave differently in each of the three sections.

3. Next, we noticed that **Cement**, **Age**, and **Super-plasticizer** have strong positive correlation with **Concrete Compressive Strength**.

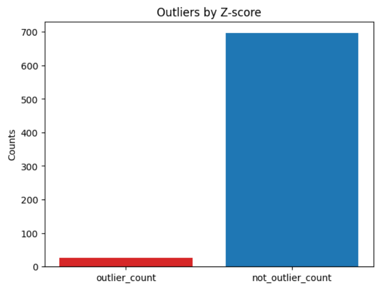
4. We observe that **Coarse Aggregate** and **Fine Aggregate** are statistically insignificant and according to the OLS model with p-values greater than 0.05 likely having no meaningful linear effect on the prediction.

5. Finally, we noticed that **Water** appears to be negatively correlated with **the Compressive Strength** of concrete with a negative coefficient of -0.1485 in the OLS model.

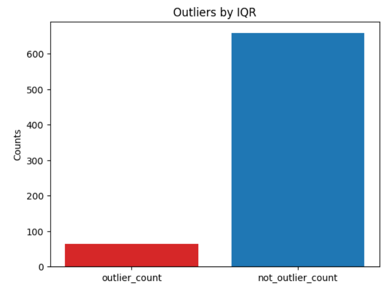
We expect some of our initial findings to line up with the results our models give in the subsequent sections.

2.2 Z-Scores and IQR methods

Our first approach was to calculate the Z-scores and identify points outside the range $[-3.5, 3.5]$. Then, we use the Inter-quartile Range method to do the same exercise.



(a) Z-Score Method



(b) IQR Method

Figure 2: Outlier Distributions

The data contains an abundance of data in the mid-ranges of certain parameters, so it may not be appropriate to shave of the ends of the curve to reduce noise. Consider the case where a positively correlated value, like **Cement**, is found outside the acceptable Z-score and IQR ranges and those rows are removed. This would remove the behavior of the strong-performing and weak-performing combinations, and the model would only serve to model average concrete compressive strength instead of highlighting uniquely high-performing or low-performing mixtures.

We validated this by running some of our models on the data with and without outliers. We noticed that the mean square error of our training data went down a lot when we removed outliers, but the mean square error of our testing data went up. This corroborates our theory that the removal of these outliers would increase overfitting of the models.

2.3 Correlation Analysis

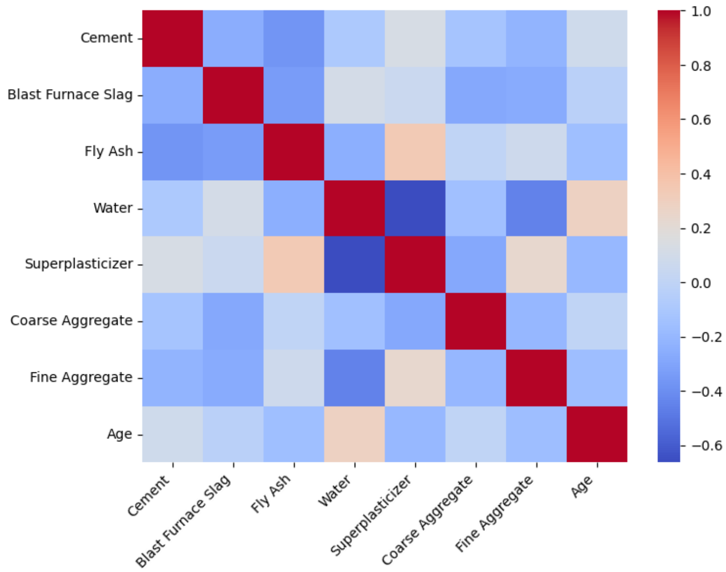


Figure 3: Predictor Correlation Matrix

3 Models and Performance

3.1 Polynomial Regression

To find the best polynomial regression model, we created multiple models while varying the degree of the polynomial. As seen in figure 4, the MSE on the test dataset is minimized when $degree = 3$. Training MSE continues to decrease after that, which suggests overfitting that is expected with higher degree polynomial regression. With $degree = 3$, we generate a model with

$$train\ MSE : 19.098 \text{ and } test\ MSE : 61.938.$$

Polynomial Regression also allows us to observe feature importance by examining feature coefficients. Observing a 2nd degree polynomial using figure 5 we can see that the first degree features are substantially more significant than the second degree features. **Water** and **Superplasticizer** appear to most significantly affect concrete compressive strength, while second degree features hold little comparable significance.

Among the second degree predictors, different predictors combined with age affect the prediction (i.e. Blast Furnace Slag$\times$Age, Fly Ash$\times$Age, Superplasticizer$\times$Age)

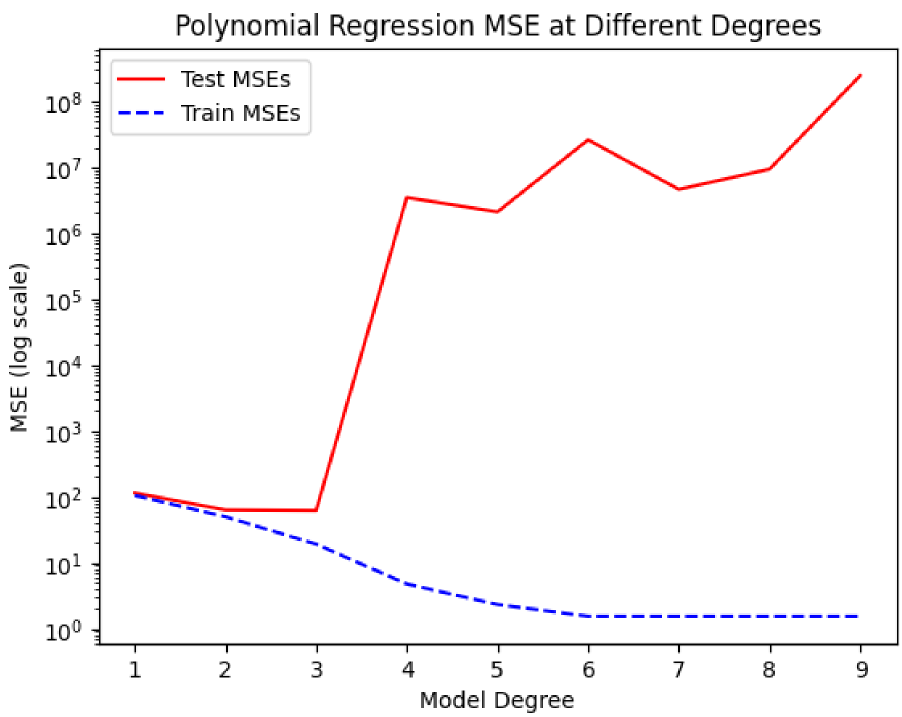


Figure 4: Graph of MSE for Polynomial Regression with varied degrees

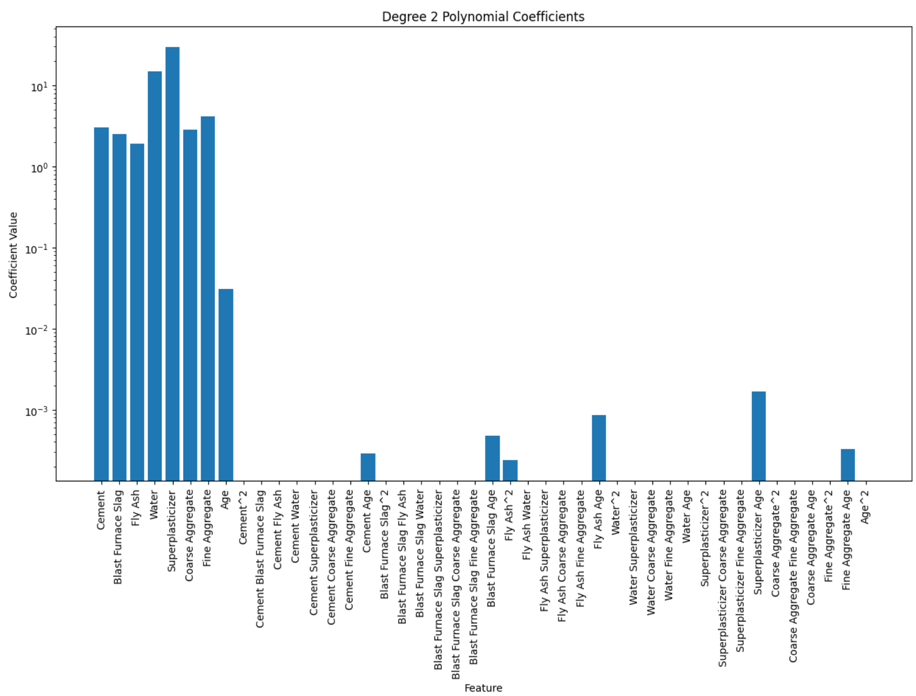


Figure 5: Feature Coefficient Values for Second Degree Polynomial Regression

3.2 Splines

The splines model has two tuning parameters: degree and number of knots. Through testing different numbers of degrees and knots for the splines model, we determine that to yield the best test MSE, the tuning parameters should be:

$$degrees = 4 \text{ and } n\_knots = 8$$

Figure 6 shows the comparison of splines models with different degrees and differing numbers of knots With our optimal splines model, we generate a model with an improved test MSE over polynomial regression

$$train\ MSE : 22.744 \text{ and } test\ MSE : 34.9852.$$

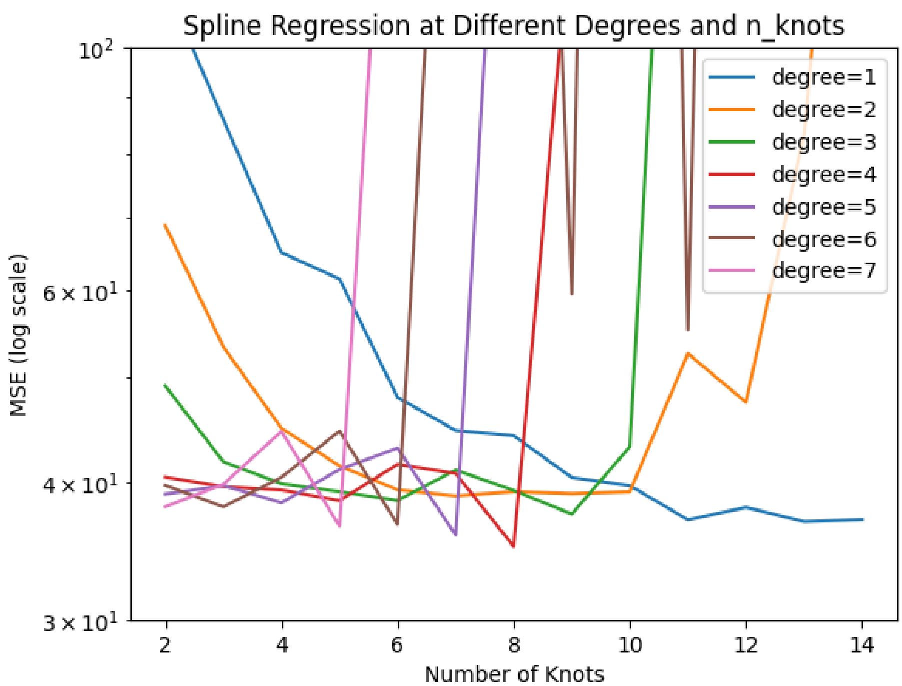


Figure 6: Comparing Various Degrees and Knots for Splines Model

Based on coefficient analysis shown in Figure 7, this splines model with degree 4 and 8 knots had **Age** and **Fine Aggregate** with the largest coefficients, indicating the importance of these features. This observation suggests that **Fine Aggregate** likely has meaningful nonlinear influence despite no linear correlation established in the OLS model.

To extend the splines model, we introduced a ridge penalty on the spline coefficients to achieve a form of smoothing. More precisely, this model is a ridge-regularized splines model. With cross-validation, we tuned the smoothing

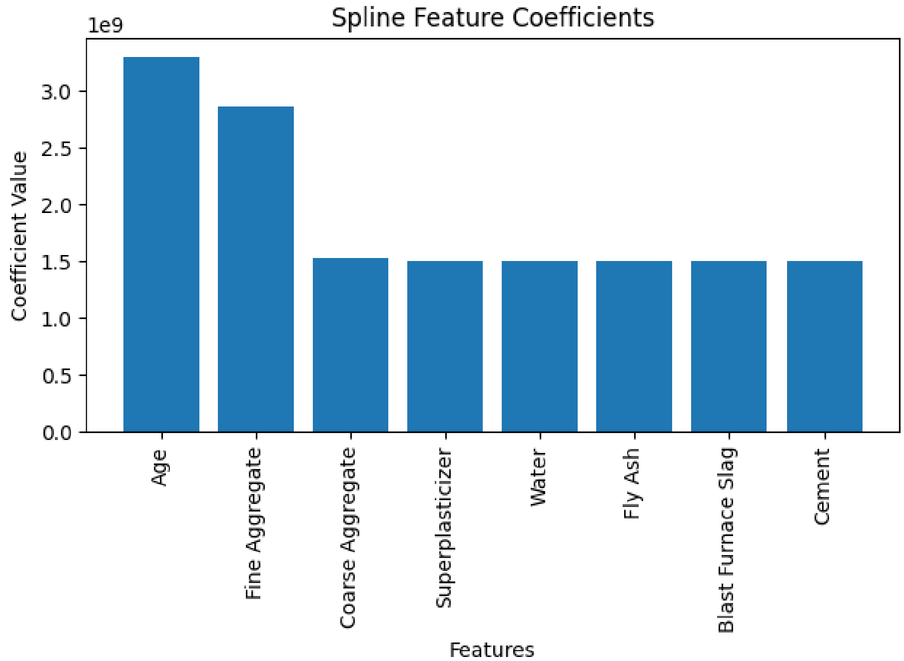


Figure 7: Coefficient Values in Optimal Splines Model

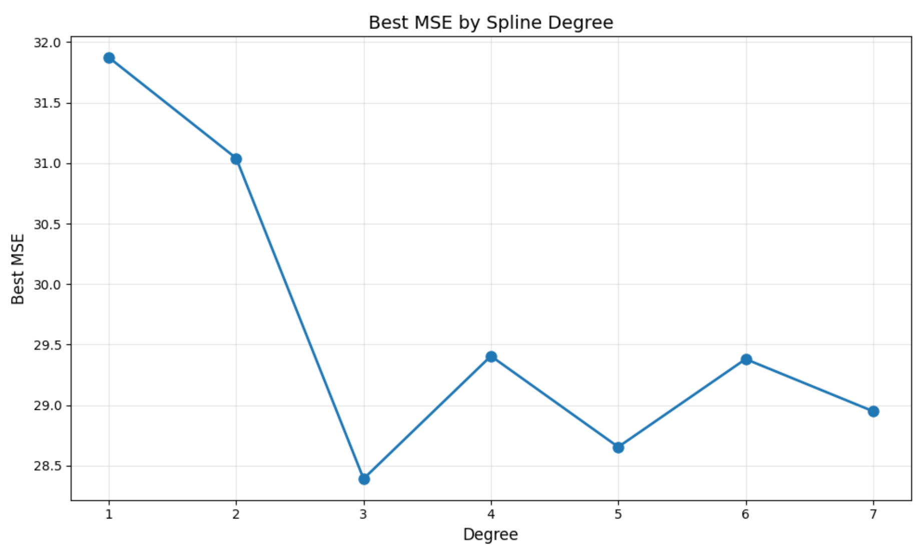


Figure 8: Degree and CV MSE Ridge-Regularized Splines Model

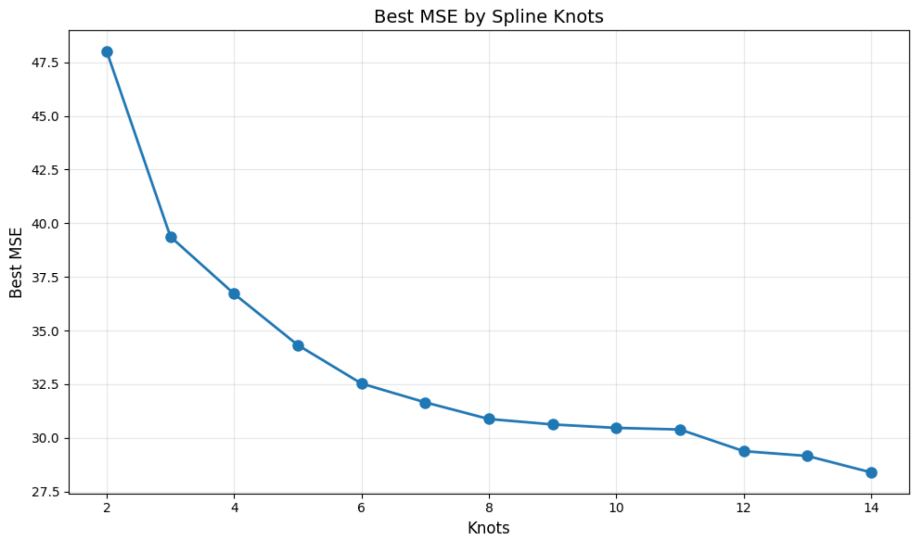


Figure 9: Knots and CV MSE Ridge-Regularized Splines Model

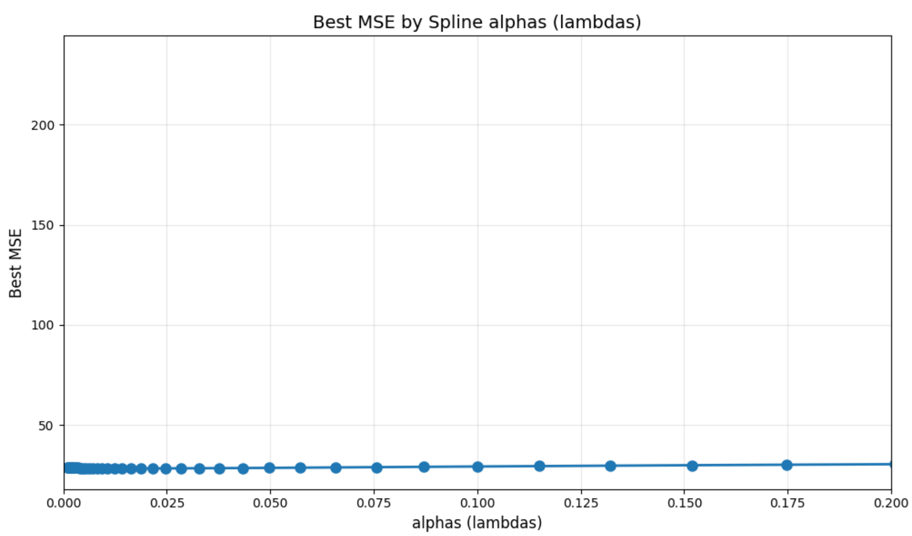


Figure 10: Lambdas and CV MSE Ridge-Regularized Splines Model

penalty alpha (lambda), the degree, and the number of knots. The results of this tuning process are illustrated in Figures 8, 9, and 10. We observe consistent improvement for both the training and the test MSE over the standard splines model. The optimal tuning parameters were found to be:

$$alpha(lambda) = 0.0187,\ degrees = 3,\ \text{and}\ n\_knots = 14$$

The improved train and test MSE for the ridge-regularized splines model over the standard splines model are:

$$train\ MSE : 20.7588\ \text{and}\ test\ MSE : 32.6033.$$

According to the coefficient analysis for the ridge-regularized splines model in Figure 11, the most important features were **Cement** and **Age**, which have the coefficients with the highest cumulative deviation from zero. The significance of these features can be easily explained. Concrete gains strength as it ages, and adding more units of cement directly impacts its final strength.

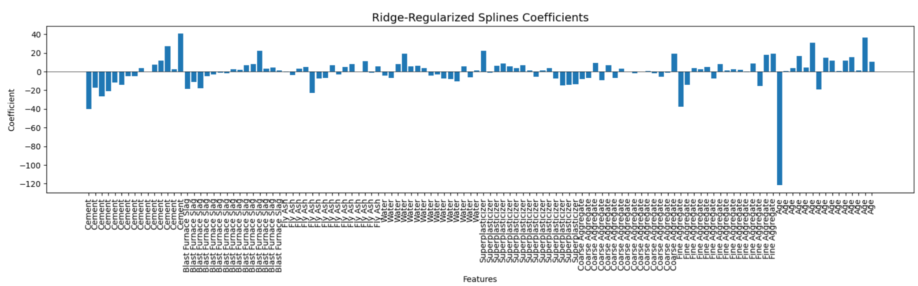


Figure 11: Coefficient Values in Ridge-Regularized Splines Model

3.3 Regression Tree

For a simple regression tree, we initially fit the tree with no max-depth and no minimum samples per leaf as shown in Figure 12. This simple regression yielded a train MSE of 1.5348 and a test MSE of 62.5701, higher than the polynomial and spline regression models. This large difference in train and test MSE is indicative of overfitting. Using cross-validation, we observed that the ideal combination of parameters was 0.0253 alpha value with a depth of 9 and a minimum of 1 sample per leaf, with a train MSE of 9.4814 and test MSE of 55.8374, still a low performer relative to the previous models. However, the pruned tree has better generalization than the simple regression tree.

$$train\ MSE : 9.4814\ \text{and}\ test\ MSE : 55.8374.$$

Like the other regression models, **Cement**, **Age**, and **Water** are the most important features, as shown in Figure 14. These features are used throughout the tree to divide the nodes.

The impact of water on compressive stress is one where more water leads to concrete being more porous and weaker, which explains the negative correlation. Therefore, when there is a node split in the tree related to water, the tree is using a specific value to split between concrete mixes that are stronger with less water content and weaker mixes with more water content.

Similar node splits exist in the tree for age and cement. For a split for age, the tree is distinguishing between concrete that has been aged for shorter durations which is weaker and stronger, longer aged concrete. For cement splits, the amount of cement in the concrete mixture impacts the strength. The tree splits on concrete mixes with more cement that generally produce stronger concrete and mixes with less cement that produce weaker concrete.

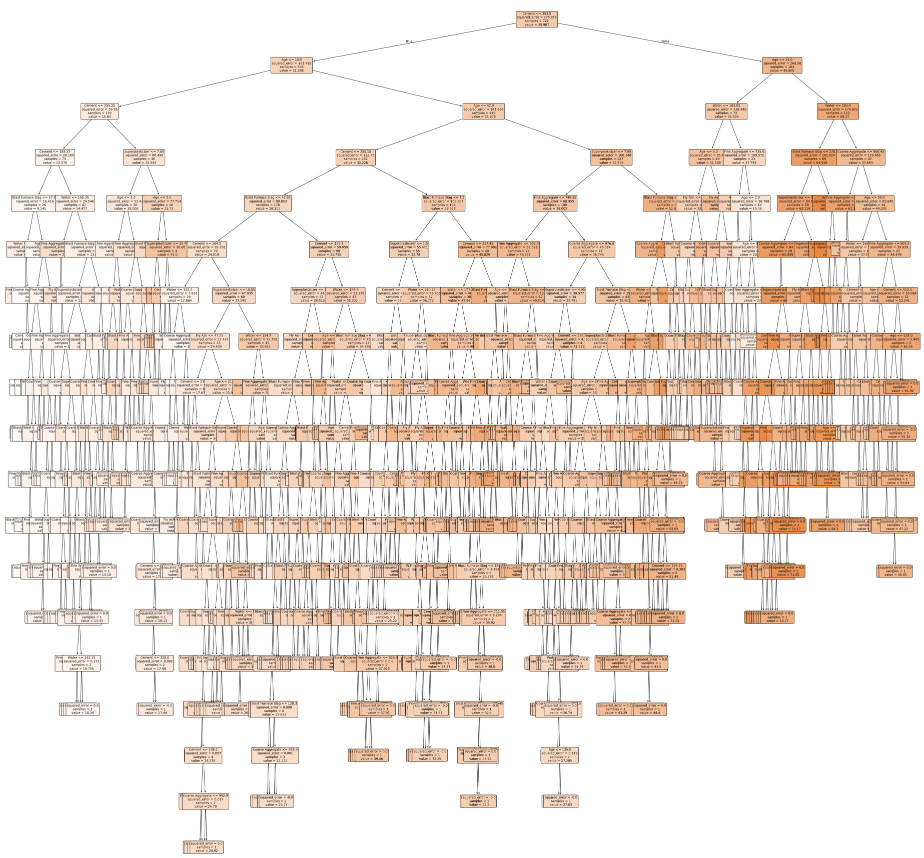


Figure 12: Full Simple Regression Tree

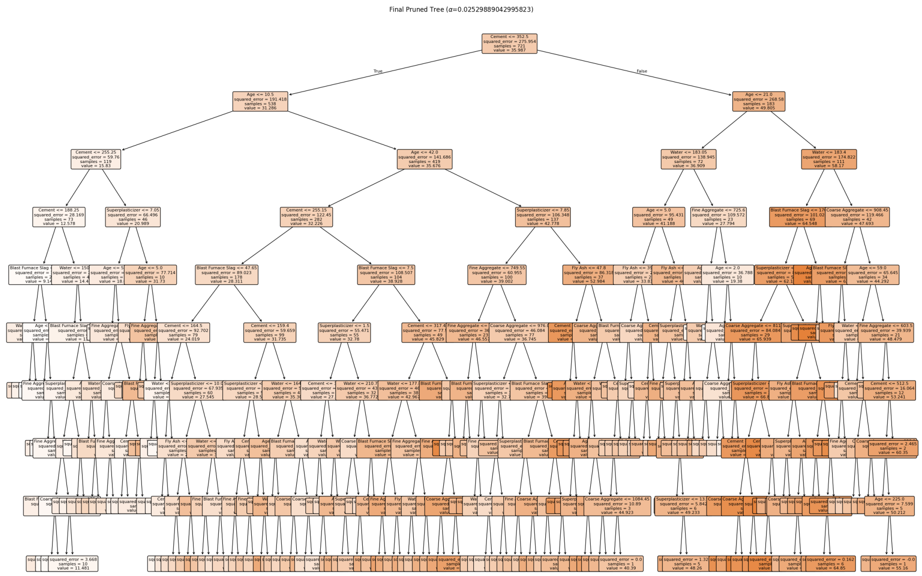


Figure 13: Pruned Tree

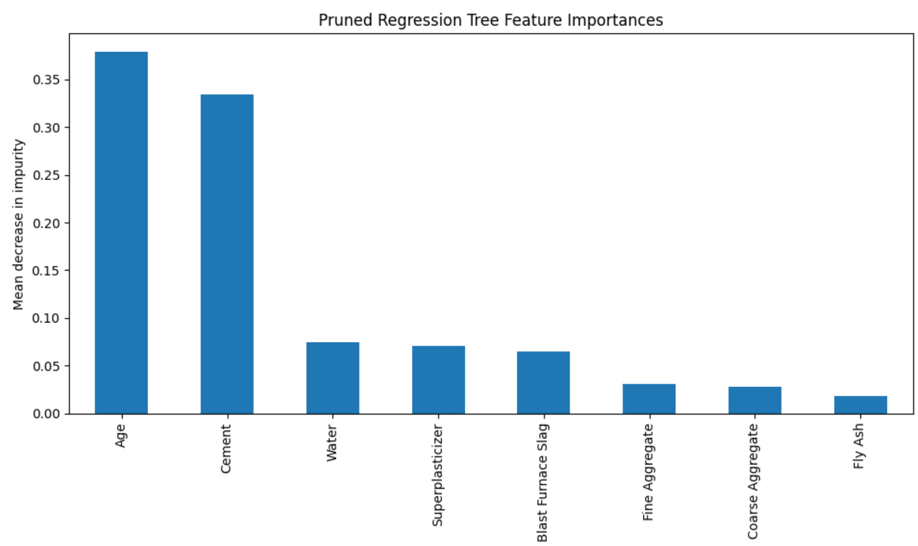


Figure 14: Feature Importance by Mean Decrease in Impurity for Pruned Tree

3.4 Random Forest

For the random forest, we performed cross-validation to explore the choice of number of estimators and max features to find the optimal number of trees and reduce correlation between trees. According to Figure 15, the optimal choice of 5 features out of the list [3, 5, 'sqrt', 'log2', 1./3., 8] of possible max features resulted in the lowest MSE. Our cross-validation process also determined the optimal number of estimators to be 300 out of the list [200, 300, 400, 500] of possible number of estimators. The resulting train MSE and test MSE are:

$$train\ MSE : 4.6006,\ OOB\ MSE : 25.6883, \text{ and } test\ MSE : 29.0374$$

This test MSE for the random forest outperforms both splines models, the polynomial model, and both the simple and pruned regression trees. Since the out-of-bag MSE is close to the the test MSE, this shows that the random forest is generalizing reasonably well.

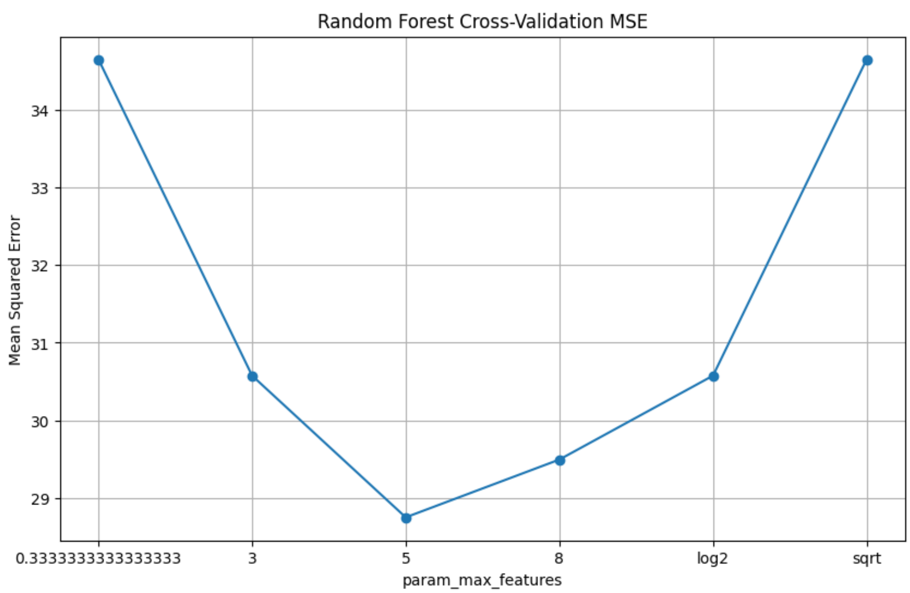


Figure 15: Comparing Parameter Limits and Transformations for Random Forest Regressions

Feature importance analysis for the random forest in Figure 16 reveals that **Age**, **Cement**, and **Water** are the most important features. The feature importance graph for the random forest is extremely similar to the one for the pruned decision tree. By the mean decrease in impurity, the random forest places slightly greater significance on the **Water** feature than the pruned tree.

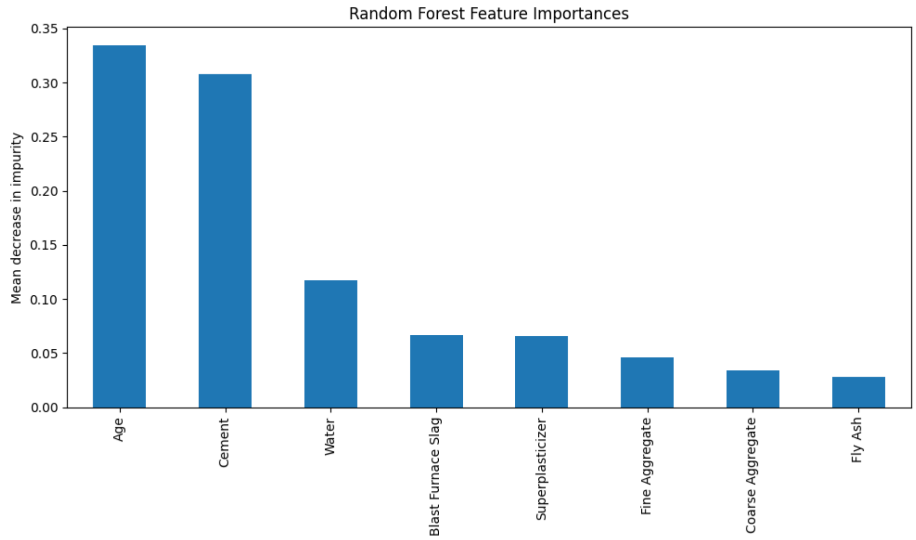


Figure 16: Feature Importance by Mean Decrease in Impurity for Random Forest

4 Conclusion

Our study has found that the most critical features that impact concrete compressive strength are **Age**, **Cement**, and **Water**. The former two features have a positive correlation and the latter has a negative correlation. The random forest model had the smallest test MSE of 29.04 as compared to the polynomial model (test MSE: 61.938), splines model (test MSE: 34.9852), ridge-regularized splines model (test MSE: 32.6033), simple regression tree (test MSE: 62.5701), and pruned regression tree (test MSE: 55.8374). Regarding the level of overfitting on the training data, the ridge-regularized splines model is less overfit with a difference between the train MSE and test MSE of around 12 as compared to a difference of 25 for the random forest model. Ultimately, for pure prediction performance, the random forest model still emerges as the optimal model.